# Sagnac Experiment

## Abstract

The goal of this experiment is to study the Sagnac effect experimentally and theoretically. Furthermore this experiment is aimed to give you experience in optical alignment of laser beam in interferometer and coupling of Gaussian beams to single mode optical fibers. The Sagnac effect is usually deemed to be a special-relativistic effect[1] produced in an interferometer when the device is rotating. Two light beams traveling around the interferometer in opposite directions require different times of flight to complete their closed path, giving rise to a phase shift proportional to the angular velocity of the apparatus. One important application of the Sagnac effect is in gyroscopes.

## Introduction

The Sagnac effect takes its name from Georges Marc Marie Sagnac, a French physicist who, in 1913, showed that the time of flight of light emitted by a source on a rotating platform and sent back by means of mirrors along a closed path was asymmetric, depending on the direction that the light travels around the apparatus. A simplified diagram of Sagnac's experiment is shown in Fig. 1. The difference in time of flight between the forward (in the sense of rotation) and backward beams is proportional to the angular velocity of the platform and appears as a phase difference measured by an interferometer. Sagnac interpreted this result as evidence against Einstein's relativity and in favor of a static luminiferous ether. It was, however, immediately and easily shown that the Sagnac effect is a fully relativistic effect.

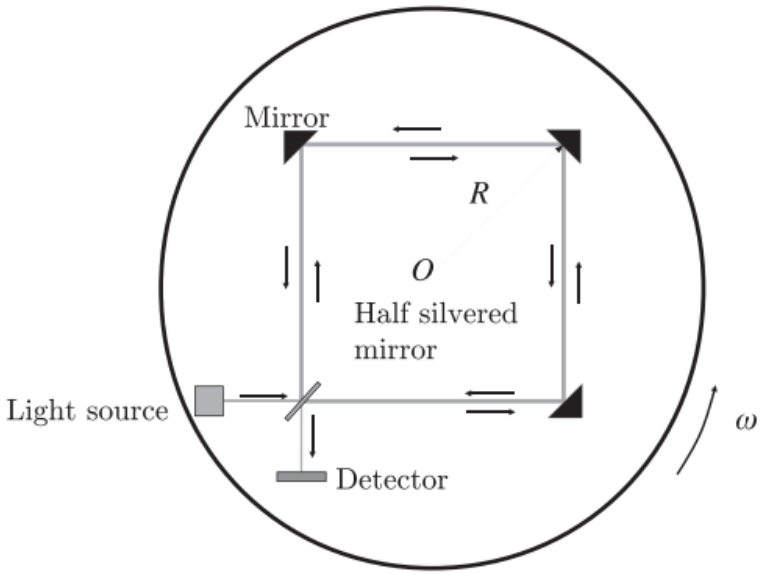



Fig. 1. Basic scheme of the Sagnac interferometric experiment. There is a turntable rotating at angular speed $\omega$. The table carries a light source, a half-silvered mirror, three more mirrors, and a detector where the interference pattern is visible. The total length of the square path is $\ell$. The arrows show the directions of the light beams.

Today, the Sagnac effect is relevant for various applications, all related to the measurement of rotation rates. On the commercial side we have gyrolasers, 3 developed since the 1970s, used on planes, ships, submarines, and missile guidance systems. The name “gyrolaser” is motivated by the fact that the devices replace old mechanical gyroscopes and use a laser to produce the counter-rotating light beams. Instead of exploiting the interference between the two opposed light rays, they measure the beat frequency of the standing wave resulting from the superposition of two waves propagating in opposite directions; that frequency is again proportional to the absolute angular velocity of the apparatus.

The most sensitive gyrolasers, or ring lasers, are used for high-accuracy geophysical measurements [2]. They sense the diurnal rotation of the Earth, the wobbles of the terrestrial axis, and tiny rotations of the laboratory hosting the instrument, due to elastic deformations of the ground caused by liquid and solid tides and also by various surface phenomena.

The extremely high sensitivity obtained in recent ring lasers is such that the Sagnac effect is now also becoming important for fundamental physics. The possibility has been suggested of using large ring lasers to detect the LenseThirring effect of general relativity on Earth, and an experiment, which goes by the acronym GINGER, is now in the development and preliminary testing phase at the National Laboratories of the Italian INFN at the Gran Sasso site.

The Sagnac effect also plays a role in the use of Global Positioning System (GPS) signals to determine the position of a receiver. A Sagnac correction is required because a rotating clock (onboard a satellite or on the surface of the Earth) loses synchronization with itself at each turn. The relevance of the relativistic effects on clocks either carried around the Earth, in the same or in the opposite sense with respect to Earth’s rotation, was experimentally verified by Hafele and Keating.

The Sagnac effect has been deeply investigated, ever since Sagnac’s experiments, as a probe of the theory of relativity and, more recently, in connection with the issue of synchronization in rotating reference frames. All of this work is well known, but it is often accompanied by a feeling that the Sagnac effect is essentially due to the presence of a rotating observer or rotating device: in short, to non-inertial motion.

# Theory

## Classical derivation of Sagnac formula

The Sagnac effect can easily be described in classical terms if one assumes that the speed of light is c with respect to a static ether. For the rotating platform described in the Introduction, one can see that if the light is going around the device in the same direction as the rotation, then it will take longer for it to reach the emission point because, meanwhile, the receiver will have moved forward by a distance $\Delta l_+ = vt_+$, where t+ is the total time of flight and v is the velocity of the emitter with respect to the ether; the geometric length of the path is l. Hence, the time of flight is read from the following relation $(l + \Delta l_+)/c = \Delta l_+/v$. For light going around in the opposite sense, the receiver will move towards the returning beam so that the path will be shorter by $\Delta l_- = vt_-$, whence $(l - \Delta l_-)/c = \Delta l_-/v$. Solving for t+ and t- gives the time of flight difference:

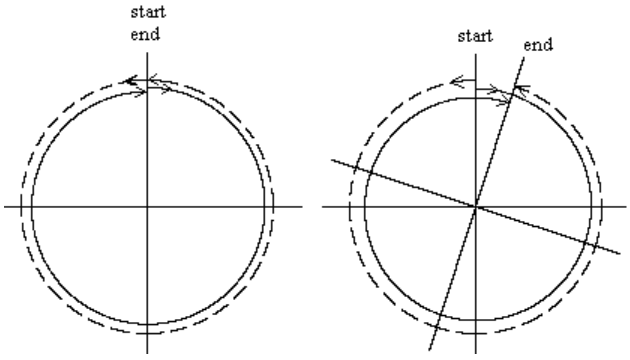



*Figure 1, Rotating loop. The pulse traveling in the same direction as the rotation of the loop must travel a slightly greater distance than the pulse traveling in the opposite direction, due to the angular displacement of the loop during the transit. As a result, if the pulses are emitted simultaneously from the "start" position, the counter-rotating pulse will arrive at the "end" point slightly earlier than the co-rotating pulse*

$$\Delta t = t_+ - t_- = \frac{2lv}{c^2 - v^2} \tag{1}$$

An equivalent deduction can be made adopting the viewpoint of the rotating observer. In the observer's reference frame, light will be expected to have speed c – v in one direction (forward) and c þ v in the other; now the path length is the same l for both (see Fig. 1). The times of flight are again different and a trivial calculation reproduces Eq. (1).

It is useful to rewrite Eq. (1) in terms of the angular velocity x of the platform. If R is the distance of the emitter/detector from the rotation axis, then we have $v = \omega R$. We simplify the geometry by assuming that the light path is a circle at radius R, so that $l = 2\pi R$. Considering that in all practical cases $v \ll c$, Eq. (1) can be written as

$$\Delta t \approx 4\frac{A \cdot \omega}{c^2} \tag{2}$$

Here A is a vector whose magnitude is equal to the area enclosed by the path of the light beams ($\pi R^2$) and is oriented perpendicular to the plane of the trajectory, while $\omega$ is the (vector) angular velocity; the

dot product projects the area into the plane of rotation. Equation (2) is the famous Sagnac formula, where the ≈ symbol is usually replaced by an equals sign, although the approximation is still present.

As mentioned above, however, the Sagnac effect does not depend on the existence of the ether. The reader is welcome to further read on the special (or general) relativistic derivation.

Interference fringes will form on the screen with a fringe displacement given by

$$\Delta\phi \approx \frac{2\pi c}{\lambda}\Delta T \tag{3}$$

where the first factor is the frequency of light. This gives the Sagnac formula:

$$\Delta\phi \approx \frac{8\pi}{\lambda c} A \cdot \omega \tag{4}$$

For N loops in a fiber equation (4) will become:

$$\Delta\phi \approx N\frac{8\pi}{\lambda c} A \cdot \omega \tag{5}$$

Where $\Delta\phi$ is the phase shift, $\lambda$ is the beam's wavelength, $c$ is the speed of light, N the number of loops in the ring in which the beam travels, $\omega$ is the angular velocity of the ring's rotation, and $A$ the area of the ring.

## Multi-Beam Interference

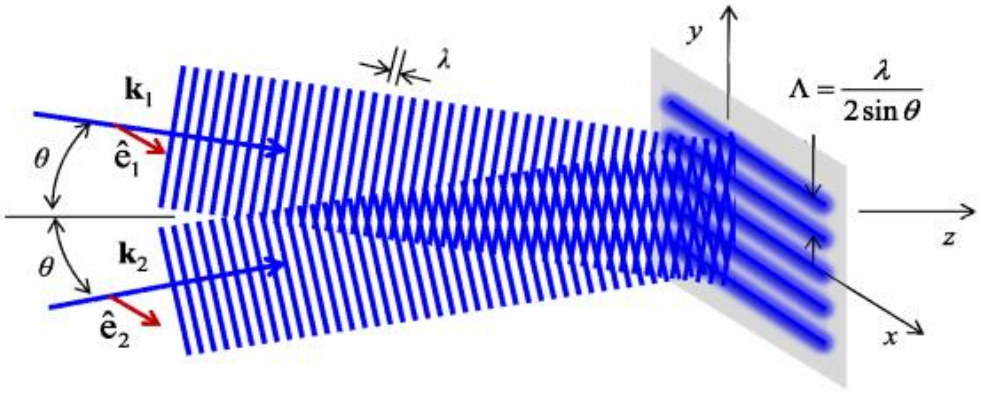
An understanding of MBI may be developed by first considering the familiar interference pattern produced by two linearly-polarized, monochromatic, plane waves derived from a common coherent source. Each plane wave may be defined by $\overrightarrow{E_i} = E_i \cos(\omega t - k_i \cdot r + \phi_i)\,\widehat{e_i}$ ,where Ei is the amplitude, $\omega$ is the frequency, ki is the wavevector, $\phi$ is the phase, and $\widehat{e_i}$ represents the linear polarization vector, and i = 1,2. The interference of the two plane waves produces a one-dimensional (1D) fringe pattern with a periodicity Λ directly proportional to the wavelength $\lambda$ and inversely proportional to the sine of the sum of the wavevector incidence angles $\theta$ with respect to the z axis as depicted in Figure 2.

The total time-independent intensity distribution $I_T(r)$ at the intersection of the two waves is expressed as the absolute square of the sum of the amplitudes $I_T(r) = |E_1(r) + E_2(r)|^2$. For the wavevector configuration depicted in Figure 1, the total intensity distribution may be given by [3]:

$$I_T(r) = I_0\left[1 + \frac{E_1E_2(\hat{e_1}\cdot\hat{e_2})}{I_0}\cos(k_0 2sin\theta y + \phi_1 - \phi_2)\right] \quad (6)$$

Where $k_0 = 2\pi/\lambda$ and a DC intensity term $I_0 = \frac{1}{2}(E_1^2 + E_2^2)$ is introduced. From Equation 6, we see that the spatial-cosine term describes a 1D periodic pattern that varies along the x axis with a periodicity of $\Lambda = \frac{\lambda}{2}sin\theta$ as depicted in Figure 2.

**Figure 2.** Two-beam interference. Interference fringes at the *x*-*y* plane with a periodicity of $\Lambda$ are formed by two linearly-polarized, monochromatic, plane waves. In this example, $\mathbf{k}_1$ and $\mathbf{k}_2$ are contained in the *y*-*z* plane at an angle $\theta$ with respect to the *z* axis.




Each minima along the y-axis (yi) in the interference fringes will be located at $y_i = \Delta\phi + \Lambda_i$, therefore the average phase shift could be calculated by the following average:

$$\Delta\phi_{Avg} = \left\langle \Delta\phi_i^{(n)} \right\rangle = \langle \frac{\Delta r_i^{(n)}}{\Lambda_i^{(n)}} \rangle \quad (7)$$

where n is the intensity pattern index of the n-th angular velocity, and i is the index of the i-th minimum.

## Fiber Optics and V-number

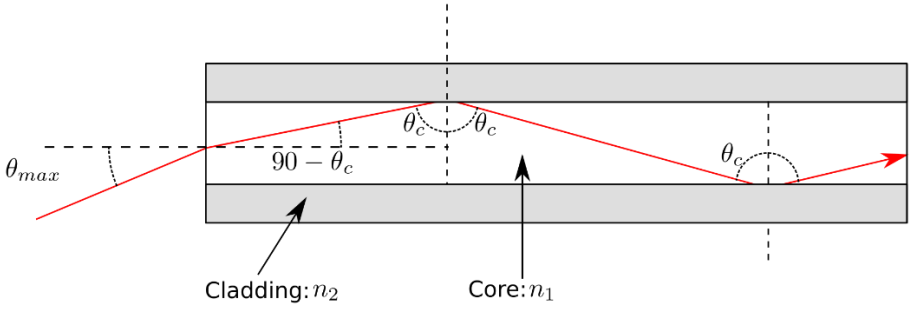



*Figure 2, A multi-mode fiber of index n1 with cladding of index n2.*

A multi-mode optical fiber will only propagate light that enters the fiber within a certain range of angles, known as the acceptance cone of the fiber. The half-angle of this cone is called the acceptance angle, θmax. For step-index multimode fiber in a given medium, the acceptance angle is determined only by the indices of refraction of the core, the cladding, and the medium:

$$nsin\theta_{max} = \sqrt{n_{core}^2 - n_{cladding}^2} = NA \tag{8}$$

where n is the refractive index of the medium around the fiber, ncore is the refractive index of the fiber core, and nclad is the refractive index of the cladding. While the core will accept light at higher angles, those rays will not totally reflect off the core–cladding interface, and so will not be transmitted to the other end of the fiber.

As a rule of thumb, the half-angle beam divergence in radians should not exceed the NA of the fiber, regardless of the core diameter. Then you should be able to get most of the light launched, assuming that is also all hits the fiber core at the interface.

The V number is a dimensionless parameter which is often used in the context of step-index fibers. It is defined as

$$V = \frac{2\pi}{\lambda} a \cdot NA = \frac{2\pi}{\lambda} a \sqrt{n_{core}^2 - n_{cladding}^2} \tag{9}$$

where λ is the vacuum wavelength, a is the radius of the fiber core, and NA is the numerical aperture.

For a wavelength of 532 nm and SMF-28e (see spec below) we have $a = 4.1\ \mu m, NA = 0.14$. So we get $V(532\ nm) = 6.78$. For comparison for 1550 nm we have $V(1550\ nm) = 2.32$.

The V number can be interpreted as a kind of normalized optical frequency. (It is proportional to the optical frequency, but rescaled depending on waveguide properties.) It is relevant for various essential properties of a fiber:

- For V values below ≈ 2.405, a fiber supports only one mode per polarization direction (→ single-mode fibers).
- Multimode fibers can have much higher V numbers. For large values, the number of supported modes of a step-index fiber (including polarization multiplicity) can be calculated approximately as $M \approx V^2/2$

- number of modes versus V number
- The V number determines the fraction of the optical power in a certain mode which is confined to the fiber core. For single-mode fibers, that fraction is low for low V values (e.g. below 1), and reaches ≈ 90% near the single-mode cut-off at V ≈ 2.405.
- There is also the so-called Marcuse equation for estimating the mode radius of a step-index fiber from the V number; see the article on mode radius.
- A low V number makes a fiber sensitive to micro-bend losses and to absorption losses in the cladding. However, a high V number may increase scattering losses in the core or at the core–cladding interface.

We can see the number of modes in an optical fiber using the following figure from [4]:



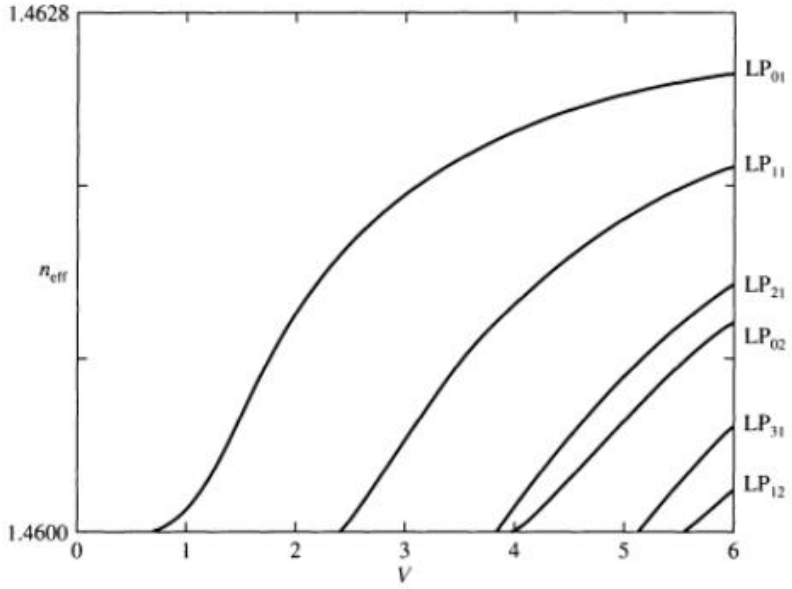



**Figure 3.12** Normalized propagation constant $n_{eff}$ as function of normalized frequency $V$ for some of the guided modes of the optical fiber, $n_{eff} = \beta/k_0$. The fiber parameters are $n_1 = 1.4628$, $n_2 = 1.4600$, and $a = 4.7$ μm.

## Excitation of LP modes

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## 10.3 EXCITATION OF LP MODES

### 10.3.1 Power Coupled to LP Mode

Now we consider the excitation problem. We consider waves impinging upon a truncated fiber tip normally as shown in Figure 10.1(*b*). The medium to the left of the truncated fiber is taken as air. For $z \le 0$, the incident wave is

$$\begin{aligned} \mathbf{E}_{\text{in}}(r,\phi,z) &= (\mathbf{e}_{\text{in}t} + \hat{\mathbf{z}}e_{\text{in}z})e^{-j\beta z} \\ \mathbf{H}_{\text{in}}(r,\phi,z) &= (\mathbf{h}_{\text{in}t} + \hat{\mathbf{z}}h_{\text{in}z})e^{-j\beta z} \end{aligned} \tag{10.23}$$

The total incident power is

$$\begin{aligned} P_{\text{in}} &= \frac{1}{2}\int_0^{2\pi}\int_0^{\infty} \text{Re}(\mathbf{E}_{\text{in}} \times \mathbf{H}_{\text{in}}^*) \cdot \hat{\mathbf{z}}\, r\, dr\, d\phi \\ &= \frac{1}{2}\int_0^{2\pi}\int_0^{\infty} \text{Re}(\mathbf{e}_{\text{in}t} \times \mathbf{h}_{\text{in}t}^*) \cdot \hat{\mathbf{z}}\, r\, dr\, d\phi \end{aligned} \tag{10.24}$$

Due to the presence of the truncated fiber, a portion of the incident wave is reflected. The reflected wave propagates in the $-z$ direction and is given by

$$\begin{aligned} \mathbf{E}_{\text{rf}}(r,\phi,z) &= (\mathbf{e}_{\text{rf}t} + \hat{\mathbf{z}}e_{\text{rf}z})e^{+j\beta z} \\ \mathbf{H}_{\text{rf}}(r,\phi,z) &= (\mathbf{h}_{\text{rf}t} + \hat{\mathbf{z}}h_{\text{rf}z})e^{+j\beta z} \end{aligned} \tag{10.25}$$

We are interested in the LP modes excited by the incident field. For simplicity, we use a single subscribe $v$, instead of two subscribes $l$ and $m$, to designate an LP mode. We assume that the modal fields $\mathbf{e}_v$ and $\mathbf{h}_v$ and the propagation constants $\beta_v$ of mode $v$ are known. Our task is to determine the amplitude of the LP mode excited in the fiber. Let the field in the step-index fiber launched by the incident wave be

$$\begin{aligned} \mathbf{E}_{\text{fb}}(r,\phi,z) &= \sum_v C_v(\mathbf{e}_{vt} + \hat{\mathbf{z}}e_{vz})e^{-j\beta_v z} \\ \mathbf{H}_{\text{fb}}(r,\phi,z) &= \sum_v C_v(\mathbf{h}_{vt} + \hat{\mathbf{z}}h_{vz})e^{-j\beta_v z} \end{aligned} \tag{10.26}$$

where $C_v$ is the mode amplitude. The summation covers all possible LP modes and the radiation mode, if excited. Power coupled into mode $v$ is

$$P_v = \tfrac{1}{2}|C_v|^2 \int_0^{2\pi}\int_0^{\infty} \text{Re}(\mathbf{e}_{vt} \times \mathbf{h}_{vt}^*) \cdot \hat{\mathbf{z}}\, r\, dr\, d\phi \tag{10.27}$$



To determine $C_v$, we make use of the continuity of the tangential components of electric and magnetic fields at the truncated fiber surface $z = 0$ and obtain

$$\mathbf{e}_{\text{in}t} + \mathbf{e}_{\text{rf}t} = \sum_v C_v \mathbf{e}_{vt} \tag{10.28}$$

$$\mathbf{h}_{\text{in}t} + \mathbf{h}_{\text{rf}t} = \sum_v C_v \mathbf{h}_{vt} \tag{10.29}$$

In the above equations, the incident field $\mathbf{e}_{\text{in}}$ and $\mathbf{h}_{\text{in}}$ and modal field $\mathbf{e}_v$ and $\mathbf{h}_v$ of the LP mode are known. But the reflected fields $\mathbf{e}_{\text{rf}}$ and $\mathbf{h}_{\text{rf}}$ and mode amplitudes $C_v$ are yet unknown. To determine $C_v$, we make a vector product of both sides of (10.28) with $\mathbf{h}_{vt}^*$, integrate the resulting expression over the entire fiber cross section, apply the orthogonality relation, and obtain

$$C_v = \frac{\int_0^{2\pi}\int_0^{\infty} (\mathbf{e}_{\text{in}t} + \mathbf{e}_{\text{rf}t}) \times \mathbf{h}_{vt}^* \cdot \hat{\mathbf{z}}\, r\, dr\, d\phi}{\int_0^{2\pi}\int_0^{\infty} \mathbf{e}_{vt} \times \mathbf{h}_{vt}^* \cdot \hat{\mathbf{z}}\, r\, dr\, d\phi} \tag{10.30}$$

But $\mathbf{e}_{\text{rf}}$ is yet unknown. Thus, expression (10.30), while exact, is not useful as it stands. For weakly guiding fibers, $n_{\text{co}}$ and $n_{\text{cl}}$ are nearly the same. Besides, the core region is much smaller than the cladding region. Thus we expect the reflected field is essentially the same as the waves reflected by a large planar boundary separating air and a dielectric medium with an index $n_{\text{cl}}$. In other words, we may approximate $\mathbf{e}_{\text{rf}t}$ as

$$\mathbf{e}_{\text{rf}t} \approx \frac{1 - n_{\text{cl}}}{1 + n_{\text{cl}}} \mathbf{e}_{\text{in}t} \tag{10.31}$$

Making use of this approximation, we obtain

$$C_v \approx \frac{2}{1 + n_{\text{cl}}} \frac{\int_0^{2\pi}\int_0^{\infty} \mathbf{e}_{\text{in}t} \times \mathbf{h}_{vt}^* \cdot \hat{\mathbf{z}}\, r\, dr\, d\phi}{\int_0^{2\pi}\int_0^{\infty} \mathbf{e}_{vt} \times \mathbf{h}_{vt}^* \cdot \hat{\mathbf{z}}\, r\, dr\, d\phi} \tag{10.32}$$

Since $\mathbf{e}_{\text{in}}$ and $\mathbf{h}_v$ are known, the integrals in (10.32) can be evaluated, and $C_v$ can be determined. Once $C_v$ is known, we have the percentage of power coupled into mode $v$. The result is

$$\begin{aligned} \eta_v &= \frac{P_v}{P_{\text{in}}} \\ &\approx \frac{4}{(1 + n_{\text{cl}})^2} \frac{\left(\int_0^{2\pi}\int_0^{\infty} \mathbf{e}_{\text{in}t} \times \mathbf{h}_{vt}^* \cdot \hat{\mathbf{z}}\, r\, dr\, d\phi\right)^2}{\left(\int_0^{2\pi}\int_0^{\infty} \mathbf{e}_{vt} \times \mathbf{h}_{vt}^* \cdot \hat{\mathbf{z}}\, r\, dr\, d\phi\right)\left(\int_0^{2\pi}\int_0^{\infty} \mathbf{e}_{\text{in}t} \times \mathbf{h}_{\text{in}t}^* \cdot \hat{\mathbf{z}}\, r\, dr\, d\phi\right)} \end{aligned} \tag{10.33}$$

### 10.3.2 Gaussian Beam Excitation

Suppose the input to the truncated fiber is an $x$-polarized Gaussian beam. We assume that the Gaussian beam aligns perfectly with the fiber axis, the beam waist is at $z = 0$, and the waist radius is $w_0$. Thus the input field at $z = 0$ is

$$\mathbf{e}_{\text{in}} = \hat{\mathbf{x}} E_{\text{in}} e^{-r^2/w_0^2} \tag{10.34}$$

$$\mathbf{h}_{\text{in}} = \hat{\mathbf{y}} \frac{E_{\text{in}}}{\eta_0} e^{-r^2/w_0^2} \tag{10.35}$$

Since the input field and the fiber geometry are rotationally symmetric, the mode excited in the fiber must also be rotational by symmetric. Therefore, we consider $\text{LP}_{0m}$ modes only. The field of $\text{LP}_{0m}$ mode in a weakly guiding step-index fiber is

$$\mathbf{e}_{0t} = \hat{\mathbf{x}} e_0(r) \tag{10.36}$$

$$\mathbf{h}_{0t} \approx \hat{\mathbf{y}} \frac{N}{\eta_0} e_0(r) \tag{10.37}$$

where $e_0(r)$ is given in (10.4).

Substituting (10.34)–(10.37) into (10.33), evaluating the integral with respect to $\phi$, we obtain an expression for the percentage power fed into the $\text{LP}_{0m}$ mode:

$$\eta_{\text{LP}0m} = \frac{P_{\text{LP}0m}}{P_{\text{in}}} \approx \frac{4N}{(1+n_{\text{cl}})^2} \frac{\left(\int_0^\infty e^{-r^2/w_0^2} e_0(r)\, r\, dr\right)^2}{\left(\int_0^\infty e^{-2r^2/w_0^2}\, r\, dr\right)\left(\int_0^\infty e_0^2(r)\, r\, dr\right)} \tag{10.38}$$

The first factor of (10.38) is essentially the *power transmission coefficient*, or *transmittance*, of plane waves incident upon a planar boundary between two dielectric media. The second factor is the *launching efficiency*:

$$T_{\text{LP}0m} = \frac{\left(\int_0^\infty e^{-r^2/w_0^2} e_0(r)\, r\, dr\right)^2}{\left(\int_0^\infty e^{-2r^2/w_0^2}\, r\, dr\right)\left(\int_0^\infty e_0^2(r)\, r\, dr\right)} \tag{10.39}$$

The numerator is an integral of the product of the input field and that of the $\text{LP}_{0m}$ mode. Often, it is referred to as the *overlap integral*.

As a simple exercise of (10.39), we approximate the modal field $e_0(r)$ by a Gaussian function with a beam waist radius $w_{\text{eff}}$,

$$e_{\text{GB}}(r) = E_0 e^{-r^2/w_{\text{eff}}^2} \qquad 0 \le r \le \infty \tag{10.40}$$

Substituting (10.40) into (10.39) and evaluating the integrals, we obtain

$$T_{\text{GB}} = \frac{4w_0^2 w_{\text{eff}}^2}{(w_0^2 + w_{\text{eff}}^2)^2} \tag{10.41}$$



Clearly, the launching efficiency depends on the two waist radii, $w_0$ and $w_{\text{eff}}$. The maximum launching efficiency is 1.0 and it is reached when $w_{\text{eff}} = w_0$.

When the exact expression for $e_0(r)$ as given in (10.4) is used in the expression (10.39), the integral has to be evaluated numerically. Marcuse showed that the launching efficiency depends on $V$ and $w_0$ [9, 10]. For the $\text{LP}_{01}$ mode with a given $V$, the launching efficiency may be maximized by varying $w_0$. Based on the numerical calculation, Marcuse obtained an empirical equation for the optimum Gaussian beam waist radius $w_0$ for the maximum launching efficiency. The empirical equation is [9, 10]

$$\frac{w_0}{a} \approx 0.65 + \frac{1.619}{V^{3/2}} + \frac{2.879}{V^6} \tag{10.42}$$

The maximum launching efficiency so obtained is plotted in Figure 10.10 as function of $V$. It is worthy noting:

1. For $V > 1.2$, the maximum launching efficiency is greater than 0.944.
2. For $V = 2.4$, very close to the cutoff of the $\text{LP}_{11}$ mode, the maximum launching efficiency is 0.9965.
3. The maximum launching efficiency reaches its peak value when $V = 2.8$.

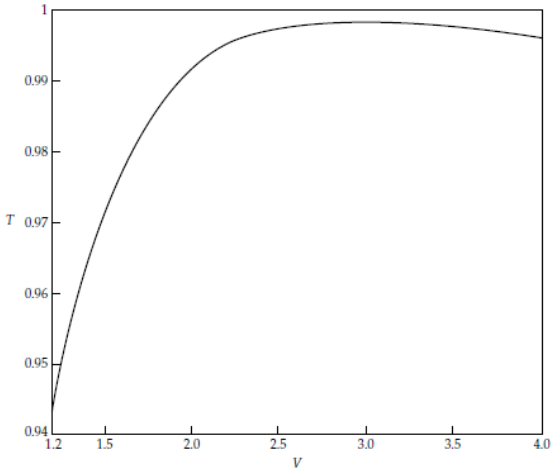



Figure 10.10 Maximum launching efficiency as a function of $V$. (From [10].)

## Numerical Aperture of an Optical System[1]

The numerical aperture of an optical system is defined as the product of the refractive index of the beam from which the light input is received and the sine of the maximum ray angle against the axis, for which light can be transmitted through the system based on purely geometric considerations (ray optics):

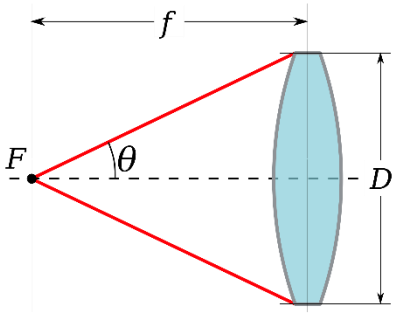



*Figure 3, Numerical aperture of thin lens*

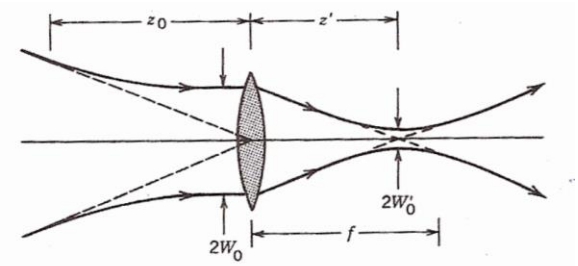
$$NA = n\sin\theta_{max} = n\sin\left[\arctan\left(\frac{D}{2f}\right)\right] \approx \frac{nD}{2f} \tag{10}$$

For the maximum incidence angle, it is demanded that the light can get through the whole system and not only through an entrance aperture.

[1] https://www.rp-photonics.com/numerical_aperture.html, https://en.wikipedia.org/wiki/Numerical_aperture

Some lenses are used for focusing collimated laser beams to small spots. The numerical aperture of such a lens depends on its aperture and focal length, just as for the collimation lens discussed above. The beam radius $w_{lens}$ at the lens must be small enough to avoid truncation or excessive spherical aberrations. Typically, it will be of the order of half the aperture radius of the lens (or perhaps slight larger), and in that case ($w_{lens}$ = D / 4 = NA · f / 2, with the beam divergence angle being only half the NA).

## Transmission of Gaussian beams through thin lenses



Focusing a beam with a lens at the beam waist.

The beam waist is defined as $W_0 = \sqrt{\frac{\lambda z_0}{\pi}}$, measurement of the beam waist provides $z_0$, for $W_0$=1 mm beam waist at 532 nm we get $z_0$ = 5.9 meter. The concentrated waist after the lens is given by [4]

$$W_0' = \frac{W}{\sqrt{1+\left(\frac{z_0}{f}\right)^2}} \approx \frac{\lambda f}{\pi W} = \frac{4\lambda f}{\pi D} \approx \frac{2\lambda}{\pi NA} \qquad (11)$$

## Optimum fiber coupling

For a wavelength of 640 nm, we have calculated V=5.6, using equation 10.42 (see image above) we get $w_0(optimum) = 3.16\,\mu m$. Using equation (11) to calculate the focal length we get f=15.5mm and NA=0.13.

For a focal length of 6 mm we get $W_0$'=1.22 µm. By using equation 10.41 substituting $w_{eff}$=4.1 µm we get the maximum launching efficiency $T_{GB}$(f=6mm) = 4*4.1^2*1.22^2/(1.22^2+4.1^2)^2=0.3 (Compared to the optimal case $T_{GB}$(f=15.5mm) = 4*4.1^2*3.16^2/(3.16^2+4.1^2)^2=0.93). Meaning if you couple more than 30% of the laser power to the fiber it is not physical with 6 mm focal length lens! In our case the focal length is 8 mm so we get $T_{GB}$(f=8mm)=0.47. So, if we use 10 mW 640 nm laser and split it to two 5mW beams, the absolute maximum intensity we can couple is 2.35 mW, one quarter of this value is acceptable.

## Single Mode Fiber SMF-28e Specifications

### Corning® SMF-28e+® Optical Fiber
**Product Information**




Corning® SMF-28e+® optical fiber is the industry leader in comprehensive single-mode fiber performance for metro and access networks. It is ITU-T Recommendation G.652.D-compliant and fully backward compatible with legacy standard single-mode fibers. SMF-28e+ fiber is built on Corning's solid foundation of quality and proven performance. Since Corning brought the first fiber to market more than 40 years ago, Corning's leadership in single-mode fiber innovation has been unparalleled.

#### *Optical Specifications*

**Maximum Attenuation**

| Wavelength (nm) | Maximum Value* (dB/km) |
|---|---|
| 1310 | ≤ 0.35 |
| 1383** | ≤ 0.35 |
| 1490 | ≤ 0.24 |
| 1550 | ≤ 0.20 |
| 1625 | ≤ 0.23 |

*Alternate attenuation offerings available upon request.
**Attenuation values at this wavelength represent post-hydrogen aging performance.

**Attenuation vs. Wavelength**

| Range (nm) | Ref. λ (nm) | Max. α Difference (dB/km) |
|---|---|---|
| 1285 – 1330 | 1310 | 0.03 |
| 1525 – 1575 | 1550 | 0.02 |

The attenuation in a given wavelength range does not exceed the attenuation of the reference wavelength (λ) by more than the value α.

**Macrobend Loss**

| Mandrel Diameter (mm) | Number of Turns | Wavelength (nm) | Induced Attenuation* (dB) |
|---|---|---|---|
| 32 | 1 | 1550 | ≤ 0.03 |
| 50 | 100 | 1310 | ≤ 0.03 |
| 50 | 100 | 1550 | ≤ 0.03 |
| 60 | 100 | 1625 | ≤ 0.03 |

*The induced attenuation due to fiber wrapped around a mandrel of a specified diameter.

**Point Discontinuity**

| Wavelength (nm) | Point Discontinuity (dB) |
|---|---|
| 1310 | ≤ 0.05 |
| 1550 | ≤ 0.05 |

**Cable Cutoff Wavelength ($\lambda_{cc}$)**
$\lambda_{cc}$ ≤ 1260 nm

**Mode-Field Diameter**

| Wavelength (nm) | MFD (μm) |
|---|---|
| 1310 | 9.2 ± 0.4 |
| 1550 | 10.4 ± 0.5 |

**Dispersion**

| Wavelength (nm) | Dispersion Value [ps/(nm-km)] |
|---|---|
| 1550 | ≤ 18.0 |
| 1625 | ≤ 22.0 |

Zero Dispersion Wavelength ($\lambda_0$): 1304 nm ≤ $\lambda_0$ ≤ 1324 nm
Zero Dispersion Slope ($S_0$): ≤ 0.092 ps/(nm²-km)

**Polarization Mode Dispersion (PMD)**

| | Value (ps/√km) |
|---|---|
| PMD Link Design Value | ≤ 0.06* |
| Maximum Individual Fiber PMD | ≤ 0.1 |

*Complies with IEC 60794-3: 2001, Section 5.5, Method 1, (m = 20, Q = 0.01%), September 2001.

The PMD link design value is a term used to describe the PMD of concatenated lengths of fiber (also known as PMDQ). This value represents a statistical upper limit for total link PMD. Individual PMD values may change when fiber is cabled.

**How to Order**
Contact your sales representative, or call the Optical Fiber Customer Service Department:
Ph: 1-607-248-2000 (U.S. and Canada)
+44-1244-525-320 (Europe)
Email: cofic@corning.com
Please specify the fiber type, attenuation, and quantity when ordering.

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#### *Dimensional Specifications*

**Glass Geometry**

| | |
|---|---|
| Fiber Curl | ≥ 4.0 m radius of curvature |
| Cladding Diameter | 125.0 ± 0.7 μm |
| Core-Clad Concentricity | ≤ 0.5 μm |
| Cladding Non-Circularity | ≤ 0.7% |

**Coating Geometry**

| | |
|---|---|
| Coating Diameter | 242 ± 5 μM |
| Coating-Cladding Concentricity | < 12 μM |

#### *Environmental Specifications*

| Environmental Test | Test Condition | Induced Attenuation 1310 nm, 1550 nm, and 1625 n (dB/km) |
|---|---|---|
| Temperature Dependence | -60°C to +85°C* | ≤ 0.05 |
| Temperature Humidity Cycling | -10°C to +85°C up to 98% RH | ≤ 0.05 |
| Water Immersion | 23°C ± 2°C | ≤ 0.05 |
| Heat Aging | 85°C ± 2°C | ≤ 0.05 |
| Damp Heat | 85°C at 85% RH | ≤ 0.05 |

*Reference temperature = +23°C
Operating Temperature Range: -60°C to +85°C

#### *Mechanical Specifications*

**Proof Test**
The entire fiber length is subjected to a tensile stress ≥ 100 kpsi (0.69 GPa).*
*Higher proof test levels available.

**Length**
Fiber lengths available up to 63.0 km/spool.

#### *Performance Characterizations*

Characterized parameters are typical values.

| | |
|---|---|
| Core Diameter | 8.2 μm |
| Numerical Aperture | 0.14<br>NA is measured at the one percent power level of a one-dimensional far-field scan at 1310 nm. |
| Effective Group Index of Refraction ($N_{eff}$) | 1310 nm: 1.4674<br>1550 nm: 1.4679 |
| Fatigue Resistance Parameter ($N_d$) | 20 |
| Coating Strip Force | Dry: 0.6 lbs. (3N)<br>Wet, 14-day room temperature: 0.6 lbs. (3N) |
| Rayleigh Backscatter Coefficient (for 1 ns Pulse Width) | 1310 nm: -77 dB<br>1550 nm: -82 dB |

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## Preparation task (to be submitted before you start the experiment)

For the following questions assume we use our 640 nm laser with a beam width of 2 mm, and the optical fiber in use is SMF-28e.

1. To achieve the highest coupling efficiency of laser light into SMF-28e fiber, what is the optimum NA of the coupling lens, assume $D_{lens}$=4 mm. What is the lens focal length?
2. What it the maximal coupling efficiency of 40x Thorlabs objective **RMS40X**? Which lens from the list below would you choose for best coupling efficiency?

| Item # | Wavelength Range | M[a] | WD | EFL | NA |
|---|---|---|---|---|---|
| **RMS4X** | Visible | 4X | 18.5 mm | 45 mm | 0.10 |
| **RMS10X** | | 10X | 10.6 mm | 18 mm | 0.25 |
| **RMS20X** | | 20X | 1.2 mm | 9 mm | 0.4 |
| **RMS40X** | | 40X | 0.6 mm | 4.5 mm | 0.65 |

3. To measure Sagnac effect we couple light to optical fibers. However, reflections from different components like beam splitter lenses or fiber holder can override the desired signal. Explain how with polarizers (and maybe using berry phase in the fibers) you can reduce this effect to a minimum.

## Minimum experimental tasks (extensions are possible)

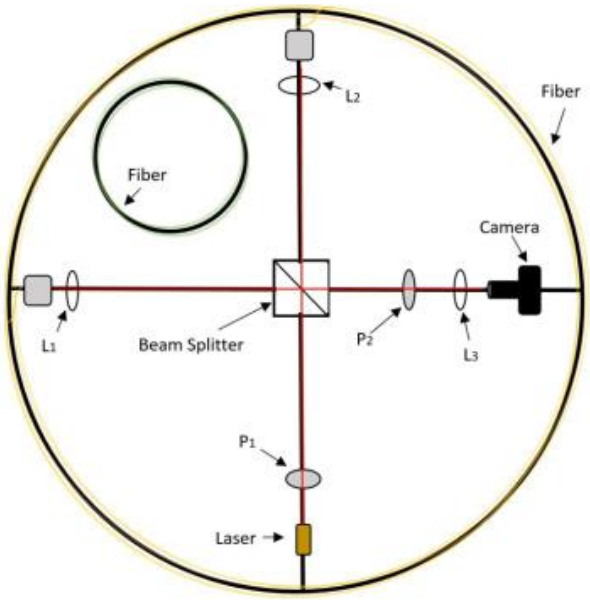



**Fig 1.** The experimental apparatus. P and L represents polarizers and lenses, respectively. The laser beam travels towards the beam splitter, thorough the first polarizer, and splits into two beams. One enters the optic fiber through L1 and exit through L2, and the second beams travel the opposite direction. The beams arrives together through the second polarizer to the camera, where the interference pattern can be seen. The system have two optical fibers – one wrapped around the large circle of the entire system, the second wrapped around a smaller circle that can be seen inside top-left quarter. The large circle is connected to an engine that can control the rotational speed of the system.

The parameters of the system are: A large circle with a radius of $57.0 \pm 0.5[cm]$, with an optic fiber wrapped around it $13 \pm 1$ times. A small circle with a radius of $15.5 \pm 0.1[cm]$ with an optic fiber wrapped around it $46 \pm 1$ times. The laser in use is one with $\lambda = 640[nm]$.

1. Measure the (linear) ratio between the voltage on the motor and its angular velocity.
2. Spin the system in different speeds and picture the interference pattern in each one.
3. Deduce the phase shift and compare with theory
4. Repeat the experiment for the off-centered small circle.
5. Evaluate the measurement errors

## References


1. A. Tartaglia and M. L. Ruggiero, "The Sagnac effect and pure geometry," Am. J. Phys. **83**(5), 427–432 (2015).

2. K. U. Schreiber and J.-P. R. Wells, "Invited review article: Large ring lasers for rotation sensing," Rev. Sci. Instrum. **84**(4), 41101 (2013).

3. G. M. Burrow and T. K. Gaylord, "Multi-beam interference advances and applications: nano-electronics, photonic crystals, metamaterials, subwavelength structures, optical trapping, and biomedical structures," Micromachines **2**(2), 221–257 (2011).

4. A. Yariv and P. Yeh, *Photonics: Optical Electronics in Modern Communications (the Oxford Series in Electrical and Computer Engineering)* (Oxford University Press, Inc., 2006).